shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >

shared_ptr< smacc2 ::introspection::SmaccState Info >

SharedPtr

SharedPtr

vector< smacc2::ISmaccState * >

SmaccStatus

map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >

recursive_mutex

atomic< StateMachineInternal Action >

map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >

map< std::string, std ::pair< std::function < std::string()>, boost ::any > >

vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >

SMRunMode

int64_t

SharedPtr

#eventsLogPub_
#stateMachinePub
_
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_

#stateMachineInfo_

#currentStateInfo_

#nh_

#timer_

#currentState_

#status_msg_

#orthogonals_

-eventQueueMutex_
-m_mutex_

-stateMachineCurrentAction

-stateCallbackConnections

-globalData_

-transitionLogHistory_

-runMode_

-stateSeqCounter_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

vector< std::shared _ptr< smacc2::ISmaccClient > >

vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >

mutex

#clients_

-clientBehaviors_

-mutex_

-stateMachine_

-smaccStateMachine_

-signalDetector_

-end_
-initialized_
-osInitialized_

-loop_rate_hz

-updatableClients_

-updatableStateElements_

-lastState_

-scheduler_

-processorHandle_

-signalDetectorThread_

-executionModel_

-statusPub_

atomic< bool >

double

vector< smacc2::ISmaccUpdatable * >

vector< std::vector < smacc2::ISmaccUpdatable * > >

atomic< int64_t >

SmaccFifoScheduler *

processor_handle

thread

ExecutionModel

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >

+stateMachine_

+parentState_

smacc2::ISmaccOrthogonal

+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()

map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >

#node_

-orthogonal_

#components_

-stateMachine_

#stateMachine_

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requireClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

std::enable_shared
_from_this< SmaccStateInfo >

map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >

vector< smacc2::introspection
::SmaccTransitionInfo >

vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >

type_info *

int

string

boost::signals2::signal
< void(const MessageType
&), optional_last_value
< typename boost::function
_traits< void(const MessageType
&)>::result_type >, int, std
::less< int >, function< void(const
MessageType &)>, typename extended
_signature< function_traits< void(const
MessageType &)>::arity, void(const MessageType
&)>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >

boost::signals2::signal
< void(const sensor_msgs
::msg::PointCloud2 &), optional
_last_value< typename boost::
function_traits< void(const sensor
_msgs::msg::PointCloud2 &)>::result
_type >, int, std::less< int >, function
< void(const sensor_msgs::msg::PointCloud2
&)>, typename extended_signature< function
_traits< void(const sensor_msgs::msg::PointCloud2
&)>::arity, void(const sensor_msgs::msg::PointCloud2
&)>::function_type, boost::signals2::mutex >

boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >

-sub_

< void(const MessageType &)>

< void(const sensor
_msgs::msg::PointCloud2 &)>

< void()>

-attitudeSub_

+staticBehaviorInfo

+stateReactorsInfo

+eventGeneratorsInfo

+transitions_

+children_

+id_

-depth_
+stateIndex_

-stateMachine_

+demangledStateName_
+fullStateName_

#owner_

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

smacc2::introspection
::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

shared_ptr< rclcpp
::Logger >

vector< std::shared
_ptr< smacc2::StateReactor > >

vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >

-currentOrthogonal

-attitudeMutex_

float

bool

SharedPtr

-topicName_

smacc2::SmaccSignal
< void(const MessageType &)>

function< void(const
MessageType &)>

optional< int >

smacc2::SmaccSignal
< void(const sensor
_msgs::msg::PointCloud2 &)>

smacc2::SmaccSignal
< void()>

-lastYaw_

+clearClouds
+minHits
+triggerClouds

-tooClose_

#stateInfo_

#logger_

#stateReactors_

#eventGenerators_

smacc2::ISmaccState

+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

+clearRangeM
+floorCutoffM
+halfAngleRad
+maxYawRateRadS
+minRangeM
+triggerRangeM

+levelFrame
+opticalFrameInput

-firstMessage_
-initialized_

+onFirstMessageReceived
+onMessageReceived_

+postInitialMessageEvent
+postMessageEvent

+queueSize

-topicName_

smacc2::client_core
_components::CpTopicSubscriber
< MessageType >

+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

#parentState_

-currentState

cl_pw4_mr::ForwardObstacle
GuardParams

+clearStreak_
+gatedClouds_
+triggerStreak_

-firstMessage_
-initialized_

-queueSize

function< void(const
sensor_msgs::msg::PointCloud2 &)>

+onFirstMessageReceived

+onMessageReceived_

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

-cosHalfAngle_

-haveAttitude_

< sensor_msgs::msg
::PointCloud2 >

+postInitialMessageEvent
+postMessageEvent

+onObstacleCleared
+onObstacleTooClose_

smacc2::client_core
_components::CpTopicSubscriber
< sensor_msgs::msg::PointCloud2 >

+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

uint64_t

array< float, 4 >

atomic< float >

atomic< int >

-params_

-subscriber_

-lastYawStampUs_

-attitudeQ_

-lastMinRange_
-yawRate_

-lastHits_

smacc2::SmaccClientBehavior

+ ~SmaccClientBehavior()
+ onEntry()
+ onExit()

cl_pw4_mr::CpForwardObstacle
Guard

+ CpForwardObstacleGuard()
+ ~CpForwardObstacleGuard()
+ onInitialize()
+ onObstacleTooClose()
+ onObstacleCleared()
+ reset()
+ isTooClose()
+ lastYawRate()
+ lastMinRange()
+ lastHits()
+ params()
- onCloud()
- onAttitude()
- levelRotation()

function< void()>

-guard_

-postCleared_
-postTooClose_

cl_pw4_mr::CbObstacleGuard

+ CbObstacleGuard()
+ ~CbObstacleGuard()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
- onTooClose()
- onCleared()